

**YEAR 10 SCIENCE - EXTENSION
CHEMISTRY REVISION**

1. Define the following.

(a) atom

(b) molecule

(c) element

(d) compound

(e) ion

(f) ionic bond

(g) covalent bond

(h) valence electron

(i) valency

2. Draw the electron shell configuration for each of the following.

(a) ${}_{16}^{32}\text{S}$

(b) ${}_{13}^{27}\text{Al}$

(c) $({}_{12}^{24}\text{Mg})^{2+}$

(d) $({}_{7}^{14}\text{N})^{3-}$

3. Using ${}^{35}_{17}\text{Cl}^{1-}$ as an example, explain the following.
- (a) Were electrons gained or lost to form the Cl^{1-} ion?
 - (b) How many electrons were gained or lost?
 - (c) In which group is Cl?
 - (d) What valency do all members of this group possess?
 - (e) Explain carefully why the ${}^{35}_{17}\text{Cl}^{1-}$ ion is stable, referring to the Octet Rule.
(HINT: Some mention of the noble or inert gases would be appropriate!)

4. Draw the electron configuration for the following compounds.

(a) magnesium sulphide (b) aluminium oxide

(c) phosphorus trichloride (d) sulphur difluoride

5. Complete the following table.

NAME	FORMULA
	CO ₂
aluminium sulphate	
dinitrogen tetroxide	
calcium hydroxide	
	(NH ₄) ₂ CO ₃
copper (II) chloride	
	NO ₂
potassium nitrate	
	ZnS
sodium phosphate	

6. Complete the following table.

ELEMENT/ION	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
$^{24}_{12}\text{Mg}$			
$(^{35}_{17}\text{Cl})^{1-}$			
$^{56}_{26}\text{Fe}$			
$(^{27}_{13}\text{Al})^{3+}$			
$^{16}_8\text{O}$			

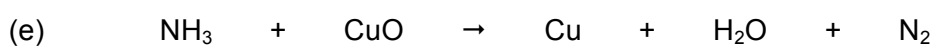
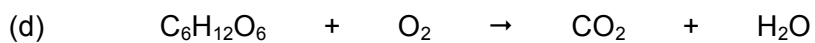
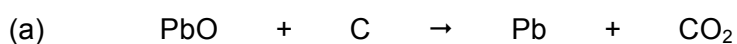
7. Explain how the following primary bonds are formed. Drawings might be very useful.

(a) Ionic bond

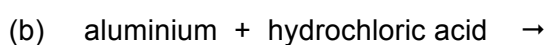
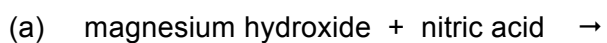
(b) Covalent bond

(c) Metallic bond

8. Balance the following equations.



9. Write **complete word equations** for the following acid reactions. Underneath each, write a **balanced equation** and an **ionic equation**.



(c) zinc carbonate + nitric acid →

(d) copper (II) oxide + sulphuric acid →

10. For the following combinations of solutions, determine if a precipitate is formed. Write a balanced ionic equation if appropriate. Write "no reaction" if appropriate.

(a) Aluminium nitrate solution and potassium carbonate solution.

(b) Copper (II) acetate solution and ammonium sulphate solution.

(c) Zinc nitrate solution and sodium hydroxide solution.

11. Calculate the formula mass for the following substances.

NAME	FORMULA	FORMULA MASS (g mol^{-1})
sodium phosphate		
iron (III) nitrate		
calcium hydrogencarbonate		
barium sulphide		
zinc acetate		

12. Convert the following to moles or mass as necessary.

(a) 36.2 g of copper (II) nitrate.

(b) 2.34 moles of aluminium hydroxide.

(b) 1.34 kg of iron (III) oxide.

(d) 1.15×10^3 moles of sulphur trioxide.

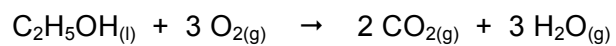
13. 27.2 g of aluminium carbonate is reacted with excess sulphuric acid. The equation for the reaction is as follows.



(a) Calculate the mass of sulphuric acid used in the reaction?

(b) What mass of carbon dioxide would be produced?

14. Ethanol burns in air according to the following equation.



If 3.85×10^2 g of ethanol burns, calculate:

(a) the mass of O_2 required.

(b) the mass of CO_2 produced.